

```
In [2]: import pandas as pd
import numpy as np
```

```
In [4]: data = pd.read_csv("CH.SC.U4CSE24156_EX-9.6_2(KNN).csv")

data.head()
```

```
Out[4]:
```

	Brightness	Saturation	Class
0	40	20	Red
1	50	50	Blue
2	60	90	Blue
3	10	25	Red
4	70	70	Blue

```
In [6]: new_brightness = float(input("Enter new Brightness: "))
new_saturation = float(input("Enter new Saturation: "))
```

```
In [8]: data['Distance'] = np.sqrt(
    (data['Brightness'] - new_brightness) ** 2 +
    (data['Saturation'] - new_saturation) ** 2
)

print(data)
```

	Brightness	Saturation	Class	Distance
0	40	20	Red	25.000000
1	50	50	Blue	11.180340
2	60	90	Blue	49.244289
3	10	25	Red	36.055513
4	70	70	Blue	39.051248
5	60	10	Red	40.311289
6	25	80	Blue	38.078866

```
In [10]: data = data.sort_values(
    by='Distance'
)

print(data)
```

	Brightness	Saturation	Class	Distance
1	50	50	Blue	11.180340
0	40	20	Red	25.000000
3	10	25	Red	36.055513
6	25	80	Blue	38.078866
4	70	70	Blue	39.051248
5	60	10	Red	40.311289
2	60	90	Blue	49.244289

```
In [12]: print("Nearest entry:")
print(data.iloc[0])
```

```
Nearest entry:  
Brightness      50  
Saturation      50  
Class           Blue  
Distance        11.18034  
Name: 1, dtype: object
```

In []:

```
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```